

Predicting & Analysing Crime in Bexley Borough

Using Machine Learning and Spatial Network Analysis to Understand Crime Hotspots and Diffusion Patterns

1. Introduction

Pattern recognition and prediction is a well-established application of machine learning – especially in the world of crime. Predictive policing tools are useful, and typically identify areas with elevated crime risk, basing judgement upon historical records, temporal patterns and local demographics. However, most do not account for the relationship between neighbouring areas, and often treat areas as spatially independent, nor do they ask whether crime in one area influences another - a phenomenon called *crime diffusion*.

This project addresses both of these drawbacks. The first notebook builds a supervised ML model to predict which Lower Super Output Areas (LSOAs) in The London Borough of Bexley will become crime hotspots in a given month. Using this data, the second notebook introduces spatial network analysis to test whether hotspot designation in an LSOA is associated with elevated hotspot probability in neighbouring areas in both one and two months' time. As touched upon briefly, this is a phenomenon known as *crime diffusion*, or alternatively known as *near-repeat victimisation*.

2. Data

The dataset comprises 59,748 street-level crime incidents recorded by the Metropolitan Police Service in Bexley Borough between March 2023 and early 2026. The data was sourced from the publicly available data.police.uk portal and consolidated from monthly source files using OpenAI Codex, which handled file merging, column standardisation, and borough filtering.

Each record includes: crime type (14 categories), month of occurrence, LSOA code and name, and latitude/longitude coordinates. The dataset covers 149 unique LSOAs across the borough. No personally identifiable information is present in the data.

Variable	Description
LSOA name	Geographic unit of analysis (149 unique areas)
Month	Month of incident (March 2023 — 2026)
Crime type	14 categories inc. violence, burglary, ASB, vehicle crime
Latitude / Longitude	Used to compute LSOA centroids and network edges
CrimeCount (derived)	Aggregated incidents per LSOA per month
Hotspot (derived)	Binary label: 1 if CrimeCount >= 75th percentile (14 incidents)

3. Notebook 01 — Hotspot Prediction

The first notebook builds a supervised classification model to predict whether an LSOA will be a crime hotspot in a given month. After aggregating incidents to LSOA-month level, hotspot labels were assigned using the 75th percentile threshold (14 or more incidents in a month). This threshold ensures a balanced class distribution while reflecting a meaningfully elevated crime burden.

A Random Forest classifier was selected for its robustness to overfitting, its ability to handle non-linear relationships, and the interpretability afforded by feature importance scores. Features included lagged crime counts, month-of-year, year, and LSOA-level crime rate history. The model was evaluated using standard classification metrics.

4. Notebook 02 — Crime Diffusion Analysis

The second notebook extends the hotspot analysis by introducing a spatial network. Each of the 149 LSOAs was represented as a node, with its position defined by the mean latitude and longitude of all incidents recorded within it. Edges were drawn between any two LSOAs whose centroids were within 1.2 km of each other, using the Haversine formula to account for the curvature of the Earth. The resulting network contains 834 edges and an average degree of 11.2 neighbours per LSOA.

The diffusion analysis asked: for each LSOA-month in which an area was a hotspot, did its network neighbours become hotspots the following month? The proportion of such cases defines the diffusion rate. Comparing this to the unconditional baseline rate yields a lift multiplier that quantifies the strength of the diffusion effect.

5. Results

Metric	Value
Total incidents	59,748
LSOAs analysed	149
Network edges	834
Avg. degree	11.2 neighbours per LSOA
Hotspot threshold	14 incidents/month (75th percentile)
Baseline hotspot rate	25.8%
Diffusion rate	30.2%
Lift multiplier	1.17x
Top diffusion hub	Bexley 015A (145 neighbour conversions)

The 1.17x lift confirms that crime diffusion is a real and measurable phenomenon in Bexley. When an LSOA becomes a hotspot, its neighbours face a meaningfully elevated probability of becoming hotspots themselves the following month. This finding is consistent with the near-repeat victimisation literature, which has documented similar patterns in burglary, vehicle crime, and violent offending across urban areas.

6. Discussion & Policing Implications

The diffusion hub analysis reveals that certain LSOAs — particularly Bexley 015A, Bexley 004B, and Bexley 004A — are disproportionately responsible for seeding hotspot formation in their neighbours. Importantly, hub status does not perfectly correlate with absolute crime volume: Bexley 015A is the top diffusion hub despite not being the highest-crime LSOA. This suggests that structural position within the spatial network — centrality and connectivity — matters as much as raw incident count.

For policing and community safety teams, this has a practical implication: resources directed at hub LSOAs may yield outsized returns by suppressing diffusion cascades before they propagate across the network. A model that ignores spatial relationships — like the standalone Random Forest — cannot identify these leverage points. The network extension adds a layer of strategic insight that purely predictive models lack.

7. Limitations

- LSOA centroids are approximated from incident mean coordinates rather than official boundary polygons, introducing small positional errors.
- The 1.2 km distance threshold is a design choice; results may vary under different thresholds.
- The analysis does not distinguish between crime types — diffusion dynamics may differ substantially between, for example, burglary and anti-social behaviour.
- Causality cannot be established from this correlational analysis; other confounders (seasonal effects, policing operations) may explain some of the observed lift.

8. AI Tool & Data Use Disclosure

In accordance with the project brief, the following discloses all artificial intelligence tools used in the production of this work, the specific role each played, and a note on dataset reuse across courses.

AI Tools Used

Tool	Provider	Role in This Project
OpenAI Codex (via ChatGPT Plus)	OpenAI	Used to engineer and clean raw Metropolitan Police street-level crime data into the consolidated CSV file used throughout this project. Tasks included merging monthly source files, standardising column formats, and filtering records to Bexley Borough.
Claude (Claude Pro / Sonnet)	Anthropic	Used to generate the spatial network visualisations in Notebook 02, including the igraph network construction, Folium interactive maps, diffusion hub scoring logic, and the majority of analytical content and written commentary in that notebook. Claude also produced the HTML project report and repository README.

What Was Not AI-Generated

The following components were produced independently without AI assistance:

- The core Random Forest model design, feature engineering decisions, and evaluation strategy in Notebook 01
- The conceptual framing of crime diffusion and near-repeat victimisation as the analytical lens
- Interpretation of results and final conclusions
- Selection of the 1.2 km distance threshold for the spatial network

Dataset Reuse Disclosure

The dataset used in Notebook 01 (01_hotspot_prediction.ipynb) — the consolidated Metropolitan Police street-level crime data for Bexley Borough — was also used as part of coursework submitted for **Dr Thompson's AI and Machine Learning class**, in which I created an interactive UI visualising the findings from notebook 1. The dataset is identical in both submissions; however, the analytical objectives, modelling approach, and outputs of this project are distinct from that prior work. This disclosure is made proactively in accordance with academic integrity expectations.

9. Conclusion

This project demonstrates that crime in Bexley Borough is not spatially independent. Hotspot LSOAs exert a measurable influence on their neighbours, with a 1.17x lift in hotspot probability observed the following month. Three LSOAs emerge as persistent diffusion hubs. These findings complement the Random Forest hotspot model by adding a spatial dimension that can inform more targeted and anticipatory policing strategies.